Backend API Documentation for Flutter Integration

This report provides a comprehensive guide to the available backend endpoints, their required request bodies, and usage instructions for the Flutter developer.

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General Information

- **Base URL**: `https://<your-ngrok-url>/api/v1`
- **Authentication**: Most endpoints require a Bearer Token in the `Authorization` header.
  - Header Format: `Authorization: Bearer <JWT_TOKEN>`
- **Content-Type**: `application/json` (unless uploading files).
- **Multipart/Form-Data**: Used for endpoints that accept images (Categories, Products, Profile).

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1. Authentication

Base Path: `/auth`

Endpoint	Method	Description	Request Body
`/signup`	POST	Register a new user	`username`, `email`, `name`, `password`, `confirmPassword`
`/login`	POST	Login to get a token	`email`, `password`

Endpoint	Method	Description	Request Body
`/forget-password`	POST	Request a password reset code	`email`
`/verify-code`	POST	Verify reset code	`resetcode` (Header: `Authorization: Bearer <reset_token>`)
`/reset-password`	POST	Set a new password	`password`, `confirmPassword` (Header: `Authorization: Bearer <reset_token>`)

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2. User Profile

Base Path: `/profile` (Requires Auth)

Endpoint	Method	Description	Request Body
`/me`	GET	Get current user details	None
`/updateMe`	PUT	Update profile info	`name` (Optional), `image` (File)
`/change-password`	PUT	Change password	`currantPassword`, `password`, `confirmPassword`
`/delete`	DELETE	Deactivate account	None

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3. Categories & Subcategories

Categories

Base Path: `/categories`

Endpoint	Method	Description	Request Body
`/`	GET	Get all categories	None
`/`	POST	Create category (Admin/Emp)	`name`, `image` (File)
`/:id`	GET	Get single category	None
`/:id`	PUT	Update category (Admin/Emp)	`name`, `image` (File)
`/:id`	DELETE	Delete category (Admin/Emp)	None

Subcategories

Base Path: `/subcategories`

Endpoint	Method	Description	Request Body
`/`	GET	Get all subcategories	None
`/`	POST	Create subcategory (Admin/Emp)	`name`, `category` (ID)
`/:id`	GET	Get single subcategory	None
`/:id`	PUT	Update subcategory (Admin/Emp)	`name`, `category` (ID)

Endpoint	Method	Description	Request Body
`/:id`	DELETE	Delete subcategory (Admin/Emp)	None

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4. Products

Base Path: `/products`

Endpoint	Method	Description	Request Body
`/`	GET	List all products	None (Use Query Params for filters)
`/`	POST	Create product (Admin/Emp)	`name`, `description`, `quantity`, `price`, `category` (ID), `subcategory` (ID), `cover` (File), `images` (Files)
`/:id`	GET	Product details	None
`/:id`	PUT	Update product (Admin/Emp)	Optional fields from POST
`/:id`	DELETE	Delete product (Admin/Emp)	None

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5. Cart

Base Path: `/cart` (Requires Auth - User Role)

Endpoint	Method	Description	Request Body
`/`	GET	Get current user's cart	None
`/`	DELETE	Clear entire cart	None
`/add`	POST	Add item to cart	`productId`
`/remove`	PATCH	Remove item from cart	`productId`
`/update`	PATCH	Update item quantity	`productId`, `quantity`

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6. Order Management

Base Path: `/order` (Requires Auth)

Endpoint	Method	Description	Request Body
`/`	POST	Create Cash Order	`address` (ID from Address endpoints)
`/`	GET	List orders	None (Users see their own, Admin sees all)
`/:id`	GET	Order details	None
`/:id/deliver`	PATCH	Mark as delivered (Admin)	None
`/:id/pay`	PATCH	Mark as paid (Admin)	None

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7. Wishlist

Base Path: `/wishlist` (Requires Auth)

Endpoint	Method	Description	Request Body
`/`	GET	Get user's wishlist	None
`/`	POST	Add to wishlist	`productId`
`/:productId`	DELETE	Remove from wishlist	None

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8. Address Management

Base Path: `/address` (Requires Auth)

Endpoint	Method	Description	Request Body
`/`	GET	Get saved addresses	None
`/`	POST	Add new address	`address: { street, city, state, zip }`
`/:addressId`	DELETE	Remove address	None

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9. Reviews

Base Path: `/reviews`

Endpoint	Method	Description	Request Body
`/`	GET	List all reviews	None
`/`	POST	Create review (User)	`comment`, `rate` (1-5), `product` (ID)
`/:id`	GET	Review details	None
`/:id`	PUT	Update review (Owner)	`comment`, `rate`
`/:id`	DELETE	Delete review (Owner/Admin)	None

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10. User Management (Admin)

Base Path: `/users` (Requires Auth - Admin Role)

Endpoint	Method	Description	Request Body
`/`	GET	List all users	None
`/`	POST	Create user	`username`, `name`, `email`, `password`, `confirmPassword`, `role`, `image` (File)
`/:id`	GET	User details	None
`/:id`	PUT	Update user	`username`, `name`, `email`, `role`, `image` (File)
`/:id`	DELETE	Delete user	None

Endpoint	Method	Description	Request Body
`/:id/changepassword`	PUT	Change user password	`password`, `confirmPassword`

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11. Global Features (Filtering & Pagination)

All "List" endpoints (Products, Categories, Orders, etc.) support the following query parameters:

- **Pagination**: `?page=1&limit=20`
- **Sorting**: `?sort=-price` (Descending) or `?sort=price` (Ascending)
- **Searching**: `?search=keyword` (Searches in name and description)
- **Field Selection**: `?fields=name,price` (Excludes fields if prefixed with `-`, e.g., `-__v`)
- **Filtering**: `?price[gte]=100&price[lte]=500`
    - Operators: `gt` (>), `gte` (>=), `lt` (<), `lte` (<=)

> *[!TIP] Always check the response status code. `200/201` for success, `400/401/403` for client errors, and `500` for server errors. Error responses usually include a `message` field describing the issue.*